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GIULIA BUZZETTI

AI and Robotics Research Engineer

ABOUT ME

PhD graduate in Industrial Innovation (Robotics & RL) specializing in learning-based control for complex, highly dynamic, and inherently unstable robotic systems. Experience designing simulation environments, reward structures, and multi-policy approaches to tackle severe instability and fall resilience. Driven by rigorous, high-cadence empirical experimentation, diagnosing sim-to-real domain gaps, and extracting critical insights from real-world hardware edge cases and experiment failures.

WORK EXPERIENCE

Industrial PhD Researcher, *Enchanted Tools*, France. **Sep 2024 – Oct 2025**

- Developed a reinforcement learning control framework for ballbot balancing, using multi-platform simulation pipelines for policy training, evaluation, and testing.
- Design of modeling strategies and implementation of a robust training pipeline for RL policies.
- Provided software support for robotic demonstrations at international events, including the Mirokai Robot Hackathon, Founders Forum Global 2025, and the AI for Good Global Summit.

Risk Data Analyst, *Banca Popolare di Sondrio*, Italy. **Jun 2021 – Oct 2022**

- Designed and automated an end-to-end KPI reporting pipeline, from process analysis and database creation to automatic ingestion of raw files into structured databases.
- Developed QlikSense dashboards for KPI computation and automated recurring report generation with NPrinting.

Tutor associated with General Mathematics II course, **Mar 2021**

Università degli Studi di Milano-Bicocca, Italy**.**

- Preparation and teaching of the exercises of the General Mathematics II course - Economics and Business Administration Bachelor's Degree.

SAP technical area internship, *Everis Italia*, Italy. **Apr 2018 - Sep 2018**

- Developed ABAP backend solutions and worked on full-stack web applications.

EDUCATION

PhD in Industrial Innovation, **Nov 2022 - May 2026**

Università degli Studi di Trento Italy - Enchanted Tools France.

- Conducted PhD research on reinforcement learning for ballbot control, addressing balancing and fall damage minimization.
- Developed RL training environments for ballbot control across multiple simulation platforms, with systematic reward analysis and comparison of control strategies.
- Published in international robotics conferences.

Master Degree in Mathematics, **Jan 2019 - Jun 2021**

Università degli Studi di Milano-Bicocca Italy.

- Thesis: Movement of an octopus' tentacle: an Optimal Control application.

Bachelor Degree in Mathematics, **2014 - 2018**

Università degli Studi di Milano-Bicocca Italy.

- Thesis: The Leap Frog method to solve wave equation.

PUBLICATIONS

- G. Buzzetti, D. Zappetti and G. Iacca, "Reinforcement Learning for Ballbot Balancing," CoDIT 2026. DOI: 10.1109/CoDIT70676.2026.11631024.
- G. Buzzetti, M. Aractingi, D. Zappetti, G. Iacca, "A Multi-policy Approach Based on Clustering for Minimizing the Damage on a Falling Ballbot," CoDIT 2025. DOI: 10.1109/CoDIT66093.2025.11321766
- G. Buzzetti, D. Zappetti, G. Iacca, "A Reinforcement Learning Method to Minimize the Damage on a Falling Ballbot," ERF 2025. DOI: 10.1007/978-3-031-76424-0_15

KEY SKILLS

- **Core Expertise**: Reinforcement Learning, Robot Modeling, Policy deployment on robot hardware, Optimal Control, Mathematical Modeling
- **Software & Frameworks**: Python, MuJoCo, Isaac Gym, MATLAB, SQL, R, C++, Git, Linux
- **Languages**: English (Fluent), French (Intermediate), Spanish (Conversational), Italian (Native)

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